

UNIVERSITÉ SALAH BOUBNIDER CONSTANTINE 3
FACULTÉ DE MÉDECINE
PREMIERE ANNÉE MEDECINE DENTAIRE

PEPTIDES

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Course outline:

1. General overview
2. Peptide bond
3. Peptides of biological interest
4. Peptide sequencing

Learning objectives:

- Define the peptide bond and identify its characteristics.
- Explain the physicochemical properties of peptides.
- Describe the biological role of certain peptides.
- Perform peptide sequencing.



PEPTIDES

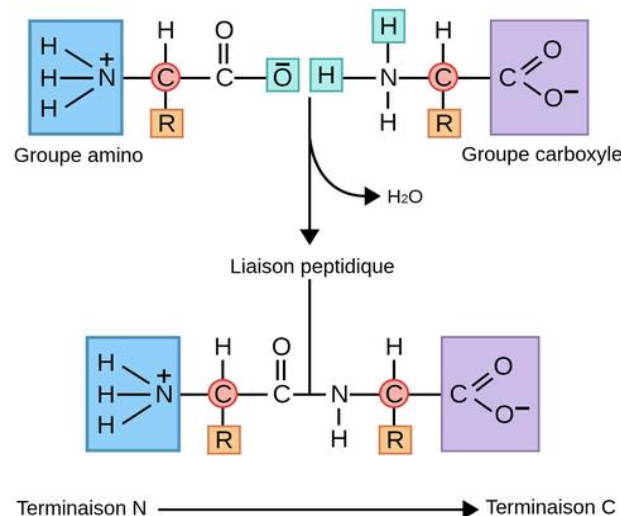
1. General overview

Peptides result from the association of two or more amino acids (AAs), linked together by covalent bonds called peptide bonds. They are classified according to the number of amino acids into:

- **Oligopeptide:** a chain of 2 to 10 amino acids
- **Polypeptide:** chains of 10 to 100 amino acids
- **Protein:** longer chains (more than 100 amino acids)

2. Peptide bond

A peptide bond covalently links amino acids through a dehydration reaction. The carboxyl group of one amino acid and the amino group of another amino acid combine, releasing a molecule of water. The resulting bond is the peptide bond. A polypeptide, regardless of its length, has one amino end (N-terminus) and one carboxyl end (C-terminus).



The reading of a peptide begins with the N-terminal amino acid (free α -NH₂) and ends with the C-terminal amino acid (free α -COOH).

2.1. Characteristics of the peptide bond (CO–NH–)

This bond is intermediate between a single and a double bond (partial double bond), which gives rise to the following properties; the peptide bond is:

- **Stable**
- **Rigid:** Free rotation around the C–N bond is not possible (important for protein conformation). However, rotation around the C α –N and C α –C bonds is possible.
- **Planar:** The six atoms involved in this bond (C α –C–O–N–H–C α) are coplanar (lie in the same plane).

2.1. Reading of a peptide chain

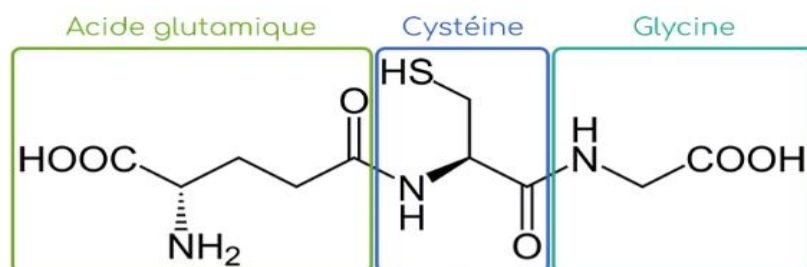
Amino acids are numbered by writing the sequence from left to right, starting from the N-terminal end toward the C-terminal end (COOH end).

The names of all amino acids are modified by adding the suffix “-yl”, except for the C-terminal amino acid.

Example: the peptide *Leu–Ser–Gly–Met* is called **leucyl-seryl-glycyl-methionine**.

3. Peptides of biological interest

- ✚ **Carnosine:** β -alanyl histidine; Carnosine is a dipeptide synthesized in the body from β -alanine and L-histidine. It results from the digestion of meat. It is found in particularly high concentrations in muscle and brain tissues.
- ✚ **Glutathione:** this tripeptide is formed by the condensation of glutamic acid, cysteine, and glycine. It is present in the cytoplasm of all our cells, where it plays a crucial role in protecting cells against oxygen attacks and its reactive forms, such as peroxides and free radicals.



Structure of glutathione

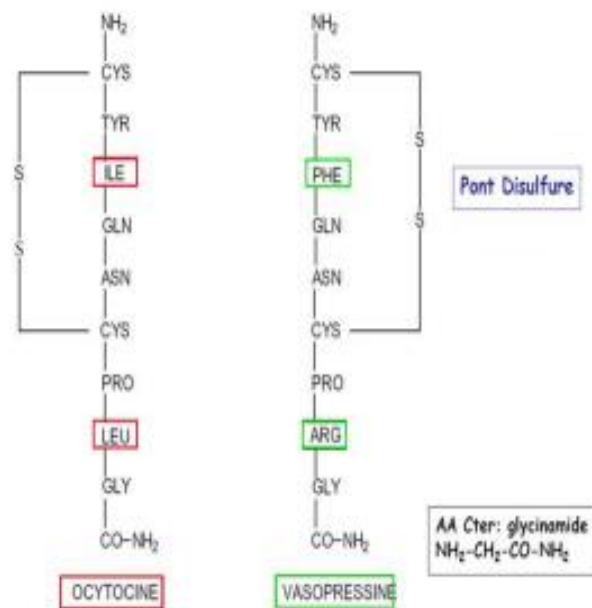
✚ Oxytocin

Oxytocin is a hormone produced by the thalamus, then stored and released by the posterior pituitary gland. It is a nonapeptide containing a disulfide bridge between Cys¹ and Cys⁶. Its main role is to promote uterine contractions during childbirth.

✚ Vasopressine

Also known as antidiuretic hormone (ADH), it is a hormone produced by the thalamus, then stored and released by the posterior pituitary gland. It is a nonapeptide that contains a disulfide bridge between Cys¹ and Cys⁶. Its main role is to promote water reabsorption in

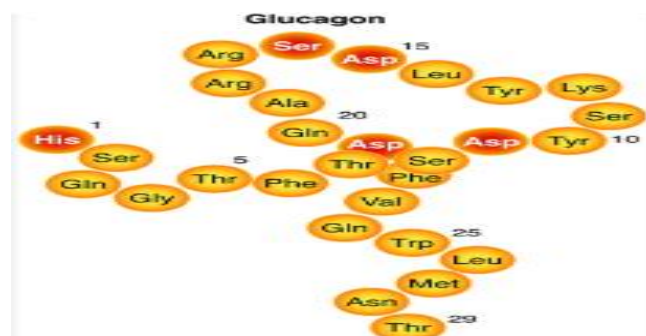
the distal renal tubules. The structural difference between vasopressin and oxytocin lies in the nature of the amino acids at positions 3 and 8.



Structural differences between oxytocin and vasopressin

+ Glucagon

Glucagon is a hyperglycemic hormone produced by the pancreas. It is composed of a single polypeptide chain consisting of 29 amino acids.



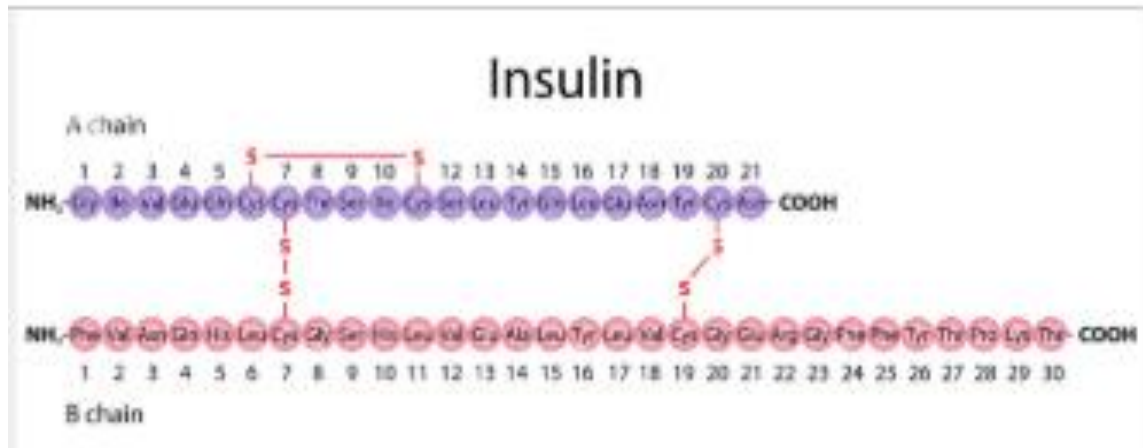
Structure of glucagon

+ Insulin

Insulin is a hypoglycemic hormone produced by the beta cells of the pancreatic islets of Langerhans. Its structure was the first to be identified among polypeptides, thanks to the research of Frederick Sanger. Insulin is composed of **two chains**:

-  **Chain A:** contains 21 amino acids
-  **Chain B:** contains 30 amino acids

The two chains are connected by **two interchain disulfide bridges** (Cys–S–S–Cys) between **Cys7–Cys28** and **Cys20–Cys39**.



Structure of insuline

2. Peptide sequencing

This is a process generally described in five steps:

1. Analysis of amino acid (AA) composition
2. Identification of N- and C-terminal residues
3. Fragmentation of the polypeptide chain
4. Arrangement of the obtained fragments and reconstruction of the peptide sequence

1. Analysis of amino acid composition:

This determination is carried out in three steps:

- Hydrolysis of the peptide bond, releasing all constituent amino acids
- Separation and identification of the released amino acids
- Quantification of the different amino acids

1.1. Hydrolysis: two approaches are possible

1.1.1. Total acid hydrolysis

Hydrolysis of peptide bonds allows the separation of the different amino acids. Total acid hydrolysis is carried out using 6 M HCl at 110°C for approximately 24 hours.

- Tryptophan is destroyed during this process
- Amide groups of glutamine and asparagine are converted into carboxylic groups: this produces glutamate and aspartate, respectively.

1.1.2. Total alkaline hydrolysis

This uses 4 M NaOH at 110°C for 24 hours.

- **Serine, threonine, cysteine, and arginine are destroyed** during this process.

Note: Total acid hydrolysis is more commonly used because it destroys fewer amino acids.

1. Identification of N- and C-terminal residues

2.1. Chemical methods:

- Sanger method (FDNB)
- Dansylation method
- Repetitive Edman degradation and treatment with hydrazine

2.1.1. Sanger method (DNFB) – N-terminal

Dinitrofluorobenzene (DNFB) binds to the NH_2 group of the first amino acid (N-terminal), followed by total hydrolysis of the peptide. Only the N-terminal amino acid is labeled as aa-DNP (yellow), then isolated and identified.

2.1.2. Dansylation method – N-terminal

Dansyl chloride (DNS-Cl) is generally used. After complete chemical hydrolysis of the peptide, the dansyl-amino acid is identified. The rest of the peptide chain is completely degraded into free amino acids.

However: the presence of lysine in the peptide can interfere with this method, because the side chain of lysine also contains an $-\text{NH}_2$ group that reacts with DNS-Cl.

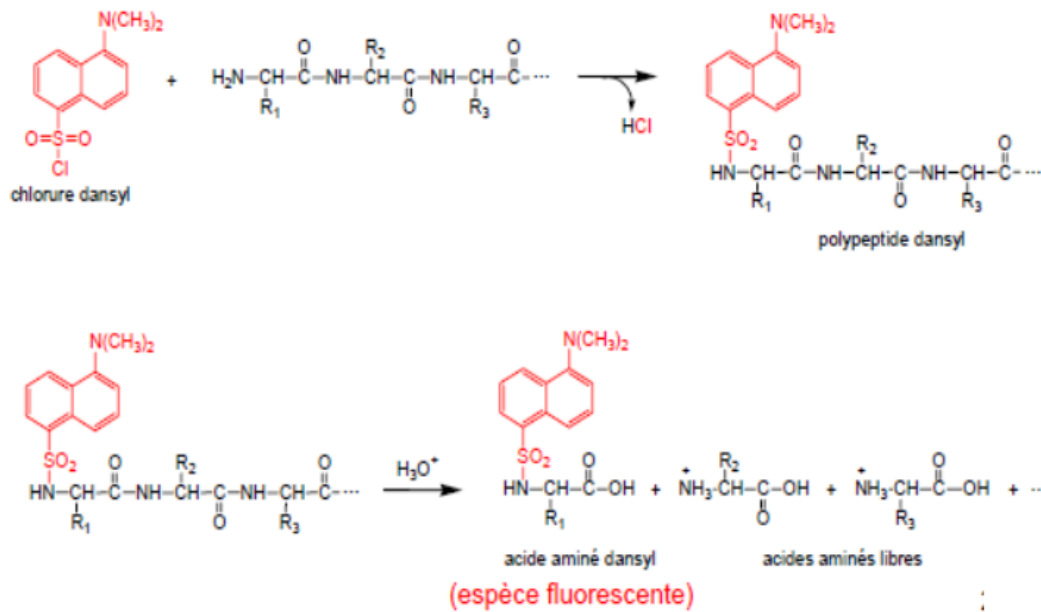


Illustration de la méthode de dansylation

2.1.3. Méthode récurrente d'Edman (N terminal) phénylthioisocyanate donne AA1-PTH. C'est la méthode la plus courante. Le groupement amino-terminal est marqué et clivé du peptide sans perturber les liaisons peptidiques entre d'autres groupements d'acides aminés.

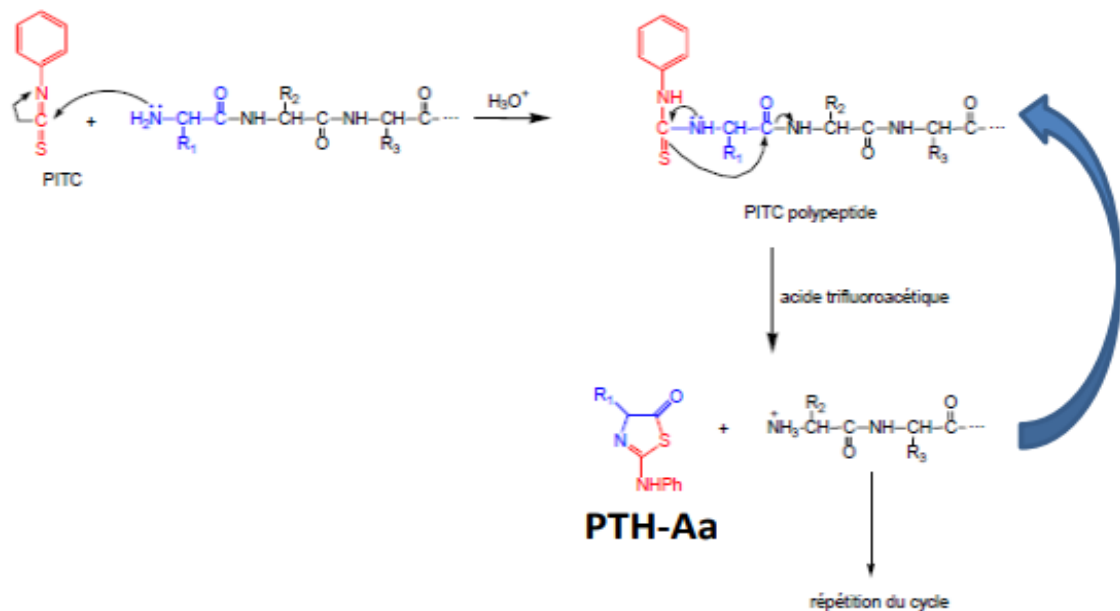
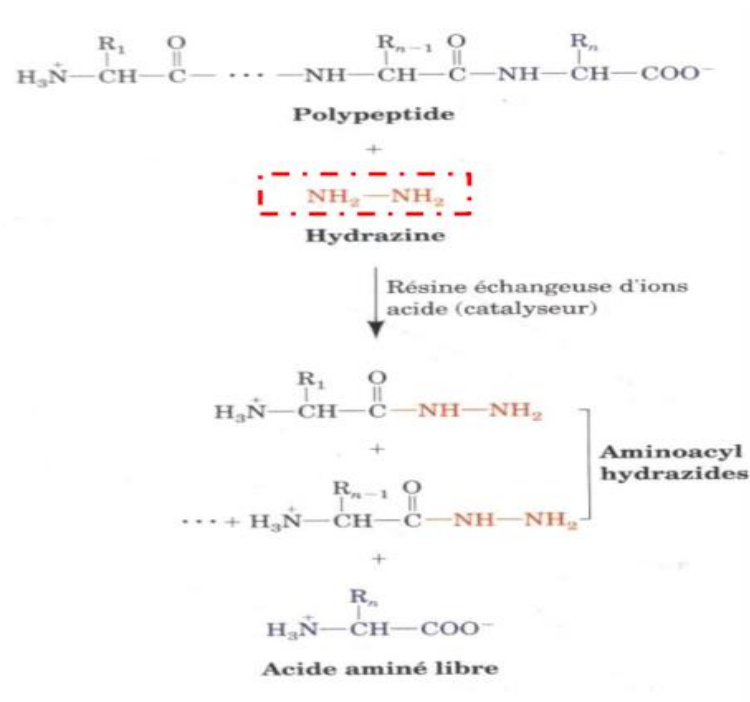


Illustration de la méthode de d'Edman

2.1.4. Traitement par l'hydrazine (C-terminal) à 100° C, qui coupe toutes les liaisons peptidiques des acides aminés et sont libérés sous forme d'aminoacyl-hydrazide à l'exception de l'acide C-terminal qui est libéré sous forme libre. Qui peut alors être identifié.



Treatment with hydrazine

2.2. Enzymatic method

2.2.1. Carboxypeptidases: These enzymes cleave the residue at the C-terminal end.

- **Carboxypeptidase A:** cleaves all residues except Pro, Arg, or Lys at the C-terminal. It is also blocked if proline precedes the C-terminal amino acid.
- **Carboxypeptidase B:** cleaves only if Arg or Lys is at the C-terminal, and not if Pro is the penultimate residue.
- **Carboxypeptidase C:** cleaves all amino acids.

3. Fragmentation of the polypeptide chain

Fragmentation is carried out by chemical or enzymatic agents that cleave the polypeptide chain at specific sites.

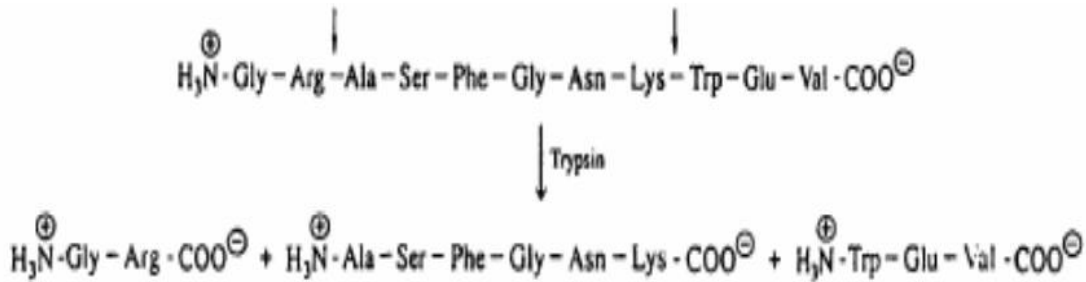
3.1. Chemical hydrolysis

- **Cyanogen bromide (BrCN):** hydrolyzes the peptide bond on the **carboxyl side of methionine**.

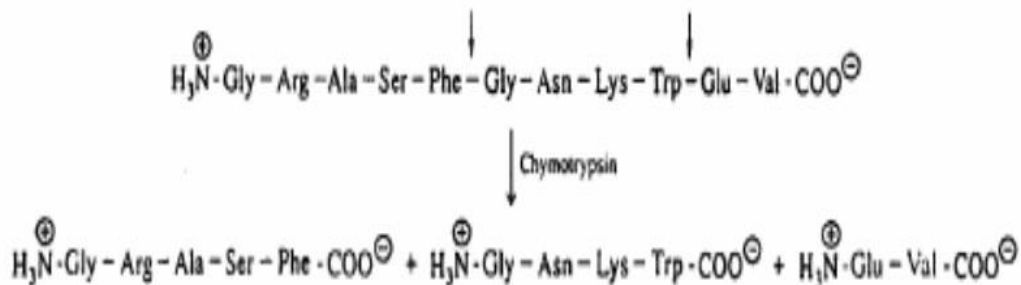
3.2. Enzymatic hydrolysis

Peptide bonds can be cleaved by proteolytic enzymes (also called proteases or peptidases).

- ✚ **Pepsin:** a gastric enzyme that specifically attacks peptide bonds involving the –NH– of Phe, Tyr, and Trp, except when Pro is on the N-terminal side.
- ✚ **Trypsin:** a pancreatic enzyme that specifically attacks peptide bonds on the carboxyl side of Lys or Arg, except when the residue following (R_n+1) is Pro.



- ✚ **Chymotrypsin:** also produced by the pancreas, it cleaves peptide bonds on the carboxyl side of aromatic amino acids (tyrosine, tryptophan, phenylalanine), except when the following residue (R_n+1) is Pro.



4. Reconstruction of the obtained fragments

Each sequenced peptide fragment corresponds to a specific part of the protein's overall sequence. The analyst then determines the correct order by assembling the fragments like a puzzle, which ultimately reveals the complete peptide sequence.

Summary table of the different chemical and enzymatic agents used in peptide sequencing

PROCESSUS	SITE	SPECIFICITE	COMMENTAIRE
1 - Clivages terminaux			
1.1. Clivage d'Edman (PTC)	côté C de l'a.a. N terminal	Rn = n'importe quel a. aminé	A l'exception des acides aminés N ter.bloqués
1.2. Carboxypeptidase A	côté N de l'a.a. C terminal	Résidu C term ($\neq$ Arg, $\neq$ Lys) Rn - 1 $\neq$ Pro	Exopeptidase
1.3. Carboxypeptidase B	côté N de l'a.a. C terminal	Résidu C term = Arg, Lys Rn - 1 $\neq$ Pro	Exopeptidase
1.4. Amino peptidase	côté C de l'a.a. N terminal	R1 = tout résidu N term sauf Pro	Exopeptidase
2 - Clivages internes			
2.1. Bromure de cyanogène (BrCN)	côté C de Rn	Rn = Met	Hautement spécifique
2.2. Trypsine	côté C de Rn	Rn = Lys, Arg Rn + 1 $\neq$ Pro	Endopeptidase hautement spécifique
2.3. Chymotrypsine	côté C de Rn	Rn = Phe, Trp, Tyr Rn + 1 $\neq$ Pro	Endopeptidase Occasionnellement Rn = Met, Asn, His, Leu
2.4. Thermolysine	côté N de Rn	Rn = Leu, Ile, Met, Ala, Val Rn - 1 $\neq$ Pro	Endopeptidase Occasionnellement Rn = Ala ou autres a. aminés
2.5. Pepsine	côté N de Rn	Rn = Leu, AA acides Phe, Trp, Tyr Rn - 1 $\neq$ Pro	Endopeptidase peu spécifique

Rn (résidu n) indique la localisation du clivage. Rn - 1 : AA à gauche, Rn + 1 : AA à droite

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